

Kinora Foot & Ankle

Physiotherapeutic Assessment Dossier

Clinical AI Orientation (Part 24)

Structured valuation workflow for RAG / AI-assisted physiotherapy consultation

Anamnesis · Inspection · Gait · Palpation · ROM · Strength · Functional & Special Tests ·
Neuro · PROMs · Return to Sport

Version 1.0 — Kinora Admin Conocimientos Upload

Complements Part 10 (Ankle) & Part 11 (Foot)

Disclaimer

*Educational resource for Kinora AI — **NOT** a substitute for licensed clinical judgment or in-person examination. Normative values are approximate and population-dependent. **Red flags — suspected fracture (Ottawa-positive), Achilles rupture (Thompson-positive), neurovascular compromise, infection, Lisfranc injury, or inability to bear weight after trauma — require urgent medical / orthopaedic evaluation.** Return-to-sport checklists do not equal clearance; athlete-specific decisions belong to the treating clinician.*

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1. Anamnesis (History)

RECORD: Chief Complaint & Pain Location Map

ID: FA-ANAM-01

Purpose: Localize primary and referred pain before exam

Key_Questions: Exact pain site? Onset timing? Traumatic vs progressive? Audible pop/snap? Able to walk? Activities that worsen/relieve?

Location_Checklist: Heel; Plantar surface; Medial arch; Lateral arch; Forefoot; Metatarsals; Hallux; Lesser toes; Hindfoot; Achilles tendon; Medial malleolus; Lateral malleolus; Dorsal foot; Plantar foot

AI_Use: Map patient free-text to checklist regions; do not invent locations not stated

Clinical_Note: Multiple sites common after sprain or in overload syndromes — rank primary vs secondary

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

RECORD: Pain Quality Descriptors

ID: FA-ANAM-02

Descriptors: Sharp/stabbing; Burning; Tightness/pulling; Electric shock; Pressure/oppression; Pain on weight-bearing; Pain with walking; Night pain; Morning pain (first-step)

Clinical_Clues: First-step morning pain → plantar fasciopathy pattern; burning/electric → neural (tarsal tunnel, radicular); night pain + trauma → fracture/infection red-flag screen

Severity: NPRS 0–10 at rest, with walking, and worst in last 24 h

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

RECORD: Injury Mechanism

ID: FA-ANAM-03

Mechanisms: Inversion; Eversion; Hyper–plantarflexion; Hyper–dorsiflexion; Overload/overuse; Running; Jumping; Cutting/COD; Direct blow

Lateral_Sprain_Pattern: PF + inversion → ATFL ± CFL; audible pop possible

Syndesmosis_Pattern: DF + external rotation (high ankle)

Medial_Pattern: Eversion / ER → deltoid ± medial clear space concern

Achilles_Pattern: Sudden push-off, felt/heard snap, immediate inability to push off

Overuse_Pattern: Gradual plantar heel / midportion Achilles / medial arch with load spike

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

RECORD: Past History & Risk Factors

ID: FA-ANAM-04

Checklist: Prior ankle sprains / CAI; Surgery; Plantar fasciitis; Fractures; Diabetes; Arthritis; Pes planus; Pes cavus; Orthotic / insole use

Red_Flags_Screen: Inability to WB 4 steps after trauma (Ottawa); deformity; neurovascular deficit; fever/redness/spreading cellulitis; sudden calf squeeze—positive Achilles rupture suspicion; night pain, unexplained weight loss, history of cancer

Diabetes_Note: Neuropathy, Charcot, delayed healing — modify loading and footwear advice

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

2. Inspection

RECORD: Weight-Bearing Inspection

ID: FA-INSP-01

Observe: Medial longitudinal arch; Transverse arch; Hindfoot alignment (varus/valgus); Hallux valgus; Claw/hammer toes; Bunions; Callosities; Edema; Erythema

Too_Many_Toes_Sign: More lateral toes visible from behind → abducted forefoot / PTTD pattern

Arch: Collapsed medial arch WB vs NWB difference suggests dynamic collapse

Skin: Callus under metatarsal heads → load transfer; erythema/warmth → inflammation/infection screen

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

RECORD: Non-Weight-Bearing Inspection

ID: FA-INSP-02

Observe: Muscle atrophy; Swelling topography; Deformities; Color; Temperature

Compare: Always bilateral — note side-to-side difference

Achilles_Contour: Gap / loss of tendon outline suggests rupture — confirm with Thompson

Atrophy: Calf wasting after prolonged unloading or tibial nerve / S1 patterns

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

3. Gait Analysis

RECORD: Gait Cycle Analysis — Foot & Ankle

ID: FA-GAIT-01

Phases: Initial contact; Midstance; Propulsion/toe-off

Observe: Pronation; Supination; Step length; Antalgic limp; Tibial rotation; Heel strike quality; Midfoot collapse; Push-off strength

Antalgic: Shortened stance on painful side — quantify visually / video if available

Propulsion_Deficit: Weak push-off → Achilles / FHL / first-ray / pain inhibition

Excessive_Pronation: Prolonged midfoot pronation — link to PTTD, plantar fascia load, medial knee stress

AI_Use: Ask patient what they notice when walking / stairs / uneven ground

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

4. Palpation

RECORD: Bony & Soft-Tissue Palpation Map

ID: FA-PALP-01

Sites: Calcaneus; Medial calcaneal tuberosity; Plantar fascia; Tibialis posterior; Fibularis (peroneal) tendons; Achilles tendon (midportion + insertional); Deltoid ligament; Lateral ligaments (ATFL, CFL, PTFL region); Base of 5th metatarsal; Metatarsal heads; Sesamoids; First ray

Ottawa Landmarks: Posterior edge/tip of lateral malleolus (6 cm); Posterior edge/tip of medial malleolus (6 cm); Base of 5th MT; Navicular — for radiograph decision after trauma

Clinical: Maximal tenderness site guides structure hypothesis; diffuse swelling less localizing acutely

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

5. Range of Motion

RECORD: Ankle Osteokinematic Norms

ID: FA-ROM-01

Dorsiflexion: $\approx 20^\circ$ (measure knee flexed and extended — gastroc vs soleus limit)

Plantarflexion: $\approx 50^\circ$

Inversion: $\approx 35^\circ$ (combined TC + STJ contribution clinically)

Eversion: $\approx 15^\circ$

Functional_DF: Weight-bearing lunge / knee-to-wall often more functional than NWB goniometry

Record: Active + passive; pain through range; end-feel; side-to-side asymmetry (cm or degrees)

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

RECORD: Hallux & First-Ray Motion

ID: FA-ROM-02

Hallux_Extension: $\approx 60\text{--}90^\circ$ needed for efficient windlass / push-off

Hallux_Flexion: $\approx 30\text{--}45^\circ$

Hallux_Limitus_Rigidus: Reduced DF of 1st MTP — pain with toe-off, dorsal osteophyte possible

First_Ray_Mobility: Assess plantar/dorsal excursion — hypermobility may impair windlass

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

6. Muscle Strength (Oxford 0–5)

RECORD: Oxford Manual Muscle Testing 0–5

ID: FA-STR-01

Scale: 0 none; 1 flicker; 2 gravity eliminated; 3 against gravity; 4 moderate resistance; 5 strong

Muscles: Tibialis anterior; Tibialis posterior; Fibularis longus; Fibularis brevis; Gastrocnemius; Soleus; Flexor hallucis longus; Extensor hallucis longus; Intrinsic foot muscles

Key_Functional: Single-leg heel-rise endurance often more informative than NWB MMT for TP/Achilles

Compare: Always vs contralateral; note pain vs true weakness

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

7. Functional Tests

RECORD: Single-Leg Stance

ID: FA-FUNC-01

Purpose: Balance and neuromuscular control

Protocol: Barefoot preferred; eyes open then closed if safe; up to 30 s

Record: Time to failure; strategies (hip strategy, toe grasp); pain; asymmetry

Clinical: Impaired after LAS/CAI; progress to foam / dual-task for athletes

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

RECORD: Heel-Raise Test

ID: FA-FUNC-02

Purpose: Achilles, triceps surae, tibialis posterior capacity

Record: Repetition count to fatigue; heel height symmetry; pain; early medial collapse (TP)

Normative_Context: Often aim ≥ 20 –25 single-leg raises for athletic RTS benchmarks (population-dependent)

Bilateral_First: If unable single-leg, document double-leg then progress

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

RECORD: Weight-Bearing Lunge Test (Knee-to-Wall)

ID: FA-FUNC-03

Purpose: Functional dorsiflexion

Normal_Guide: ≈ 10 –12 cm toe-to-wall distance commonly cited (side-to-side asymmetry critical)

Method: Knee touches wall while heel stays down; maximize distance

Clinical: DF deficit after immobilization/sprain predicts altered landing mechanics

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

RECORD: Navicular Drop Test

ID: FA-FUNC-04

Purpose: Estimate midfoot pronation / arch collapse from sitting subtalar neutral to standing

Normal_Guide: Often $< \approx 10$ mm cited; higher values suggest greater drop (interpret with symptoms)

Limitations: Measurement error; use as part of cluster not alone

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports

Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

RECORD: Windlass Test

ID: FA-FUNC-05

Purpose: Plantar fascia / windlass mechanism

Positive: Reproduction of plantar medial heel / fascial pain with passive hallux extension (WB or NWB variants)

Clinical: Supports plantar fasciopathy hypothesis with first-step pain history

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

RECORD: Jack Test

ID: FA-FUNC-06

Purpose: Integrity of windlass / ability of hallux DF to raise medial arch

Positive Finding: Failure of arch rise with hallux DF suggests impaired windlass (e.g., PTTD / midfoot dysfunction context)

Use With: Windlass test, FPI, single heel rise

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

RECORD: Foot Posture Index (FPI-6)

ID: FA-FUNC-07

Purpose: Classify standing foot posture

Categories: Highly supinated; Supinated; Neutral; Pronated; Highly pronated

Clinical: Posture ≠ pathology alone — combine with symptoms and load history

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

RECORD: Star Excursion Balance Test (SEBT)

ID: FA-FUNC-08

Directions_Core: Anterior; Posteromedial; Posterolateral (most used clinically)

Normalize: Reach distance as % of stance-leg length

Clinical: Dynamic balance / reach asymmetry after ankle injury

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

RECORD: Y-Balance Test (YBT)

ID: FA-FUNC-09

Purpose: Clinical SEBT variant — highly useful for RTS screening

Record: Anterior / posteromedial / posterolateral reaches; composite; LSI %

Asymmetry: Anterior reach asymmetry thresholds researched in athletic populations (~4 cm discussed in literature — use current protocols)

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

RECORD: Hop Test Battery

ID: FA-FUNC-10

Includes: Single hop; Triple hop; Crossover hop; Side hop; Timed hop

Metric: Limb Symmetry Index (LSI) — commonly target $\geq 90\%$ for RTS discussions

Safety: Only when pain, swelling, and strength allow; not in acute unprotected sprain

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

RECORD: Deep Squat Observation

ID: FA-FUNC-11

Observe: Arch collapse; Pronation; Dynamic valgus; Ankle DF mobility; Heel lift

Clinical: Quick functional screen of kinetic chain contribution to foot load

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

8. Special Tests

RECORD: Anterior Drawer Test (Ankle)

ID: FA-SPEC-01

Structure: ATFL — anterior talar translation

Positive: Increased translation ± sulcus vs contralateral

Timing: More reliable when acute swelling/guarding settles

Evidence Note: Variable Sn/Sp across studies — use in cluster with history + palpation

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

RECORD: Talar Tilt Test

ID: FA-SPEC-02

Structure: Primarily CFL (inversion stress); interpret with ATFL involvement

Positive: Increased inversion talar tilt vs other side

Clinical: Higher-grade lateral complex involvement

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

RECORD: Kleiger Test (External Rotation)

ID: FA-SPEC-03

Structure: Deltoid ligament / medial clear space stress; also used in syndesmosis context

Positive: Medial pain (deltoid) or syndesmotic pain depending on force/location

Caution: Correlate with squeeze test and AITFL tenderness for high ankle

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

RECORD: External Rotation Stress Test (Syndesmosis)

ID: FA-SPEC-04

Structure: Distal tibiofibular syndesmosis

Positive: Pain over AITFL / syndesmosis with DF-ER

Clinical: High ankle sprain — longer recovery than ATFL sprain typically

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

RECORD: Squeeze Test (Syndesmosis)

ID: FA-SPEC-05

Structure: Syndesmosis / interosseous membrane irritation

Positive: Distal syndesmotic pain when compressing mid-tibia/fibula

Use: Cluster with ER stress + palpation of AITFL

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

RECORD: Thompson (Simmonds) Test

ID: FA-SPEC-06

Structure: Achilles tendon continuity

Positive: Absent / markedly reduced PF of foot with calf squeeze → rupture suspicion

Action: Urgent orthopaedic pathway if positive — do not delay for 'wait and see' loading

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

RECORD: Royal London Hospital Test

ID: FA-SPEC-07

Structure: Achilles tendinopathy (insertional/midportion context)

Concept: Tenderness that lessens with active DF (tendon moves under finger) supports tendinopathy vs other local pain

Use_With: Arc sign, load history, VISA-A

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

RECORD: Arc Sign (Achilles)

ID: FA-SPEC-08

Structure: Midportion Achilles tendinopathy

Positive: Swelling/area of thickening that moves with PF/DF of ankle

Clinical: Helps distinguish moving tendon pathology from static paratenon/other local swelling

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

RECORD: Ottawa Ankle Rules

ID: FA-SPEC-09

Purpose: Decide need for ankle radiograph after acute trauma

Criteria_Summary: Pain in malleolar zone AND any of: bone tenderness posterior edge/tip lateral malleolus; bone tenderness posterior edge/tip medial malleolus; OR inability to bear weight 4 steps both immediately and in ED/clinic

AI_Use: If Ottawa-positive features → recommend imaging/medical evaluation; do not clear fracture by chat alone

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

RECORD: Ottawa Foot Rules

ID: FA-SPEC-10

Purpose: Decide need for foot radiograph after midfoot trauma

Criteria_Summary: Pain in midfoot zone AND any of: bone tenderness at base of 5th MT; bone tenderness at navicular; OR inability to bear weight 4 steps both immediately and in ED/clinic

Clinical: Lisfranc / navicular / 5th MT base injuries — do not miss

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

RECORD: Hallux Limitus / Rigidus Screen

ID: FA-SPEC-11

Assess: 1st MTP DF ROM; pain with grind; dorsal osteophyte prominence

Limitus: Reduced motion with pain — still some motion

Rigidus: Advanced stiffness / near-ankylosis pattern

Functional: Impaired toe-off; may alter lateral column loading

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

9. Neurological Screen

RECORD: Dermatome / Myotome / Reflex Screen (L4–S1)

ID: FA-NEURO-01

Dermatomes: L4 medial leg/foot; L5 dorsum foot / 1st web space; S1 lateral foot

Myotomes: L4 dorsiflexion; L5 hallux extension; S1 plantarflexion

Reflexes: Achilles (S1–2); Patellar (L3–4) for proximal comparison

Peripheral_Nerves: Deep/superficial fibular; tibial; sural; medial/lateral plantar — map if burning/paresthesia

Red_Flag: Progressive motor loss, saddle anesthesia, or bilateral deficits → urgent medical pathway (not foot-local only)

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

10. Outcome Questionnaires (PROMs)

RECORD: FAAM — Foot and Ankle Ability Measure

ID: FA-PROM-01

Domains: ADL subscale; Sports subscale

Scoring: % scale; higher = better function

Use: Ankle/foot-specific PROM for progress and RTS discussions

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

RECORD: FFI — Foot Function Index

ID: FA-PROM-02

Domains: Pain, disability, activity limitation themes (version-dependent)

Use: Foot-focused disability tracking

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

RECORD: LEFS — Lower Extremity Functional Scale

ID: FA-PROM-03

Scale: 0–80; higher better

Use: Broad lower-extremity function when problem not purely foot-isolated

MCID_Note: ≈ 9 points commonly cited — population-dependent

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

RECORD: AOFAS — American Orthopaedic Foot & Ankle Score

ID: FA-PROM-04

Note: Clinician-rated components historically common in ortho literature; know limitations vs patient-reported measures

Use: Context when reviewing surgical outcomes literature; prefer FAAM/LEFS for patient-centered tracking when possible

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

11. Return to Sport Criteria

RECORD: Recommended Return-to-Sport Checklist (Foot & Ankle)

ID: FA-RTS-01

Pain: $\leq 2/10$ with sport-specific tasks

Swelling: No meaningful edema with load progression

ROM: Full functional ROM vs other side (esp. DF)

Strength: $\geq 90\%$ vs uninjured side (MMT/dynamometry/heel-rise capacity)

Heel_Raises: ≈ 25 single-leg heel raises without pain (common athletic benchmark — individualize)

Y_Balance: $\geq 90\%$ symmetry (and no large anterior asymmetry per protocol)

Hop_Tests: $\geq 90\%$ LSI across hop battery when indicated

Sport_Tasks: Run, sprint, jump, COD pain-free

24h_Rule: Complete sport-specific training without symptom flare in following 24 hours

Psych: Athlete confidence / fear of re-injury addressed

AI_Use: Present as criteria to discuss with clinician — not automatic clearance

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

12. Evidence & Bibliography

RECORD: Core Clinical Practice Guidelines & Texts

ID: FA-EVID-01

CPG_Ankle: APTA/JOSPT Clinical Practice Guideline — Lateral Ankle Ligament Sprains (2021)

CPG_Plantar: APTA/JOSPT Clinical Practice Guideline — Plantar Heel Pain / Plantar Fasciitis (2023)

Measures: Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus themes)

Texts: Magee — Orthopedic Physical Assessment (7th ed.); Dutton — Orthopaedic Examination; Brukner & Khan's Clinical Sports Medicine (6th ed.)

Imaging_Rules: Ottawa Ankle Rules; Ottawa Foot Rules

Use_In_RAG: Prefer CPG recommendations for acute LAS and plantar heel pain pathways; cite document names in Fuente lines

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.

RECORD: How This Dossier Relates to Parts 10–11

ID: FA-EVID-02

Part_10: Kinora_Ankle_AI_Orientation.pdf — anatomy, ligaments, pathologies, rehab depth

Part_11: Kinora_Foot_AI_Orientation.pdf — foot bones, arches, forefoot pathology depth

Part_24: This dossier — end-to-end assessment workflow for AI/clinician orientation

Workflow: Use Part 24 to structure exam; pull Parts 10–11 for structure-specific detail and rehab progressions

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. APTA/JOSPT Lateral Ankle Ligament Sprains CPG (2021). APTA/JOSPT Plantar Heel Pain / Plantar Fasciitis CPG (2023). Dutton M. Orthopaedic Examination, Evaluation, and Intervention. Clinical Measures of Musculoskeletal Foot and Ankle Assessment (international consensus). Ottawa Ankle Rules / Ottawa Foot Rules primary literature.